

HW5

in slides

	S	R	Q
5-(1)	1	0	1
	0	0	0
	0	1	0
	1	1	0

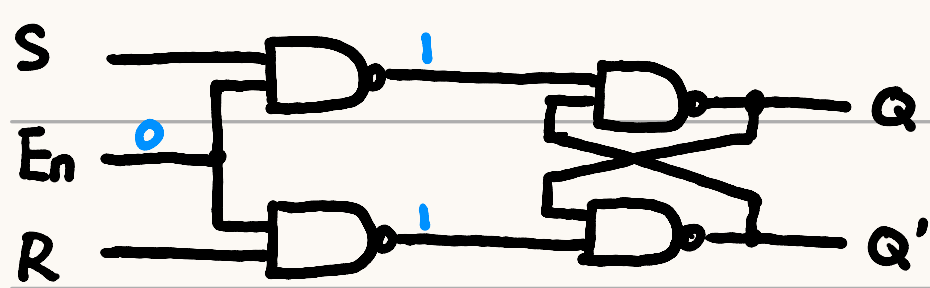
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The SR Latch uses positive feedback to create memory

5-2 problems: When S, R are both 1, both Q and Q' are 0, violating the complementary rule. Return S=R=0 also causes a race condition.

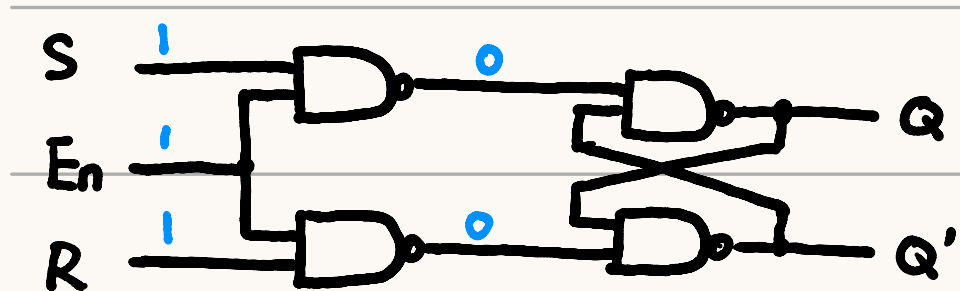
need flip-flop: The latch is an asynchronous circuit. As long as S or R is high, the output reacts immediately to any input change, complicating time control in large systems.

5-(3) $E_n = 0$ S=1 R=1



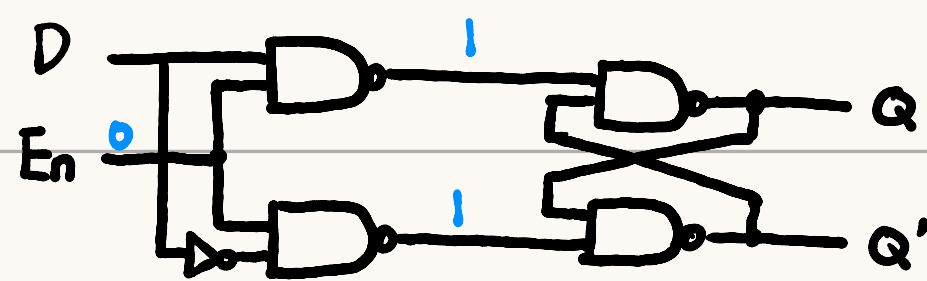
according SR Latch with NAND gates, when inputs are both 1, the output keeps.

$E_n = 1$ S=1 R=1



according SR Latch with NAND gates, when inputs are both 0 and return S=R=1, it will causes race condition

5-(4) $E_n = 0$ D=x



according SR Latch with NAND gates, when inputs are both 1, the output keeps.

5-(5)

Clk	D	Y	Q
0	1	X ₁	X ₁
0	0	X ₂	X ₂
1	0	0	X
1	1	1	X

D flip-flop: The circuit is used to store one bit of data. It use D Latches in series to achieve edge-triggered behavior.

Clk: It acts as a synchronization signal. It

controls: at Clk=1, the master Latch reads D, at clk=0, the slave reads Y locked by master.

5-(6) flip-flop input equation

$$J_A = B \quad K_A = x'B$$

$$J_B = x' \quad K_B = A \oplus x$$

get it from the figure

state equation

$$A(t+1) = BA' + (x+B')A = A'B + AB' + Ax$$

$$B(t+1) = x'B' + (Ax + A'x')B = x'B' + xAB + x'A'B$$

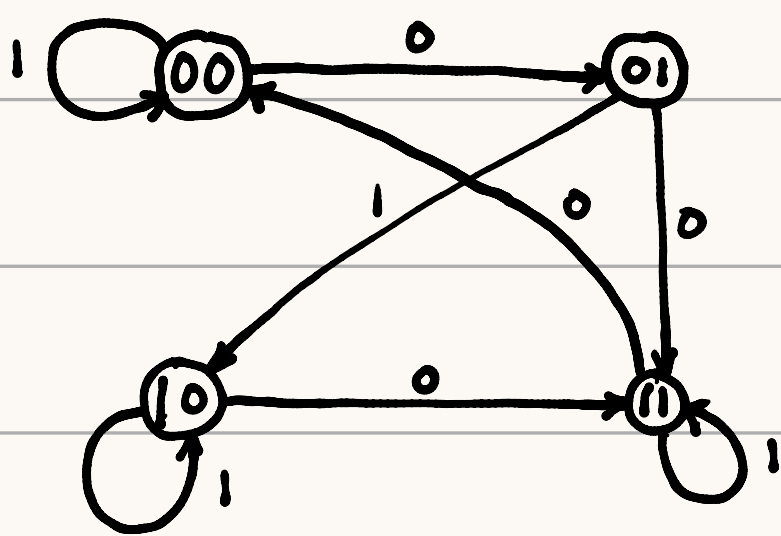
plugin the J and k in to $D = JQ' + K'Q$ function

state table

A_0	B_0	x	A_1	B_1
0	0	0	0	1
0	0	1	0	0
0	1	0	1	0
0	1	1	1	0
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1
1	1	1	0	1

A_0 and B_0 are initial state, when x input, the state of A_0 and B_0 will change to A_1 B_1

state diagram



the state table and diagram are same fundamentally.

5-(7)

$$T_A = Bx \quad T_B = x$$

$$y = AB$$

$$A(t+1) = Bx A' + (Bx)' A = Ax' + B'A + A'Bx$$

$$B(t+1) = xB' + x'B$$

The circuit functions as a 2-bit synchronous up counter with a control input.

in Textbook

$$5.3 \quad Q(t+1) = JQ' + K'Q$$

$$\begin{aligned} Q'(t+1) &= (JQ' + K'Q)' \\ &= (JQ')' (K'Q)' \end{aligned}$$

$$= (J' + Q)(K + Q')$$

$$= J'K + J'Q' + KQ$$

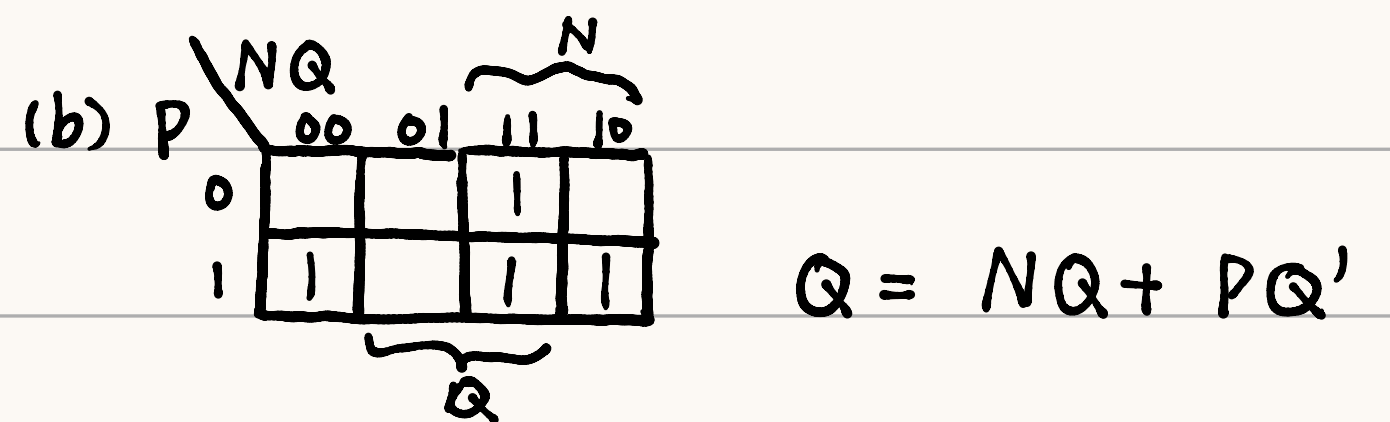
$$= J'KQ + J'KQ' + J'Q' + KQ$$

$$= (1 + J')KQ + (1 + K)J'Q'$$

$$= KQ + J'Q'$$

5.4 (a)

P	N	$Q(t)$	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



$$5.7 \quad S = x \oplus y \oplus Q$$

$$C = xy + xQ + yQ$$

$$Q(t+1) = xy + xQ + yQ$$

x	y	Q	$Q(t+1)$	S
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

